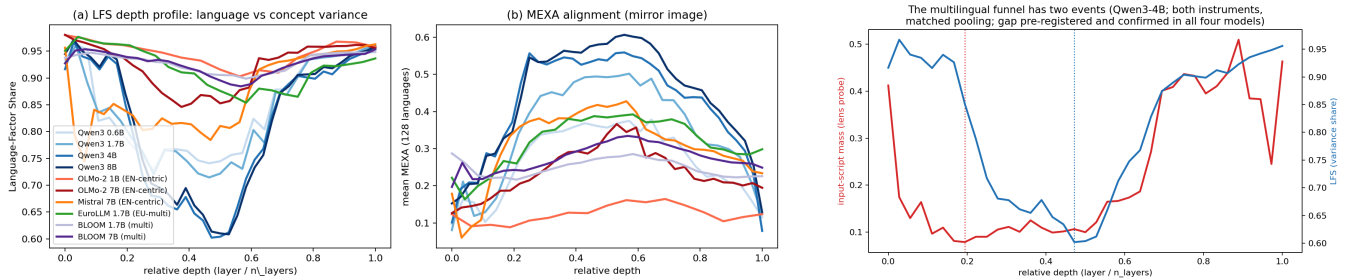


Aims 1 and 2 update: measuring the language component in multilingual LLMs

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17 models (0.6B–9B) · 122–128 languages · 31 pre-registered predictions, 21 passed · 4/4 blind prospective test · 7 self-caught errors, all documented



The **Language-Factor Share** (LFS): per-layer variance share owned by language identity; a direct operational measure of §3.1.1’s “how strong the language component is.”

The depth profile has **two events**: script abandoned early, concepts converge mid; pre-registered, confirmed twice.

Ten findings

1. **§3.1.1, measured**: shared-space depth dips in the middle layers and recovers, and varies 0.04–0.34 across models; a measurable, model-dependent quantity.
2. **Two-event depth profile**: surface script drops out early (~ 0.1 depth); meanings converge mid (~ 0.4). New structure, twice confirmed.
3. **Family separates more than scale**: between-family spread at fixed size dwarfs within-family scaling slope; Qwen3-1.7B out-converges every other lab’s 7B.
4. **Coverage $\neq$ convergence**: BLOOM (deliberately 46-language) has the flattest concept space; its alignment lives in the embedding layer (surface form, not meaning), invariant from 1.7B to 7B.
5. **“Thinks in English” is resource-graded and fading**: a multilingual latent factor out-predicts English for 113 to 124 of 127 languages by model (124 for most) (+0.08 after confound correction); English’s best-donor share falls 45 $\rightarrow$ 25/127 as Qwen3 scales.
6. **The mean hides the tail**: mid-resource languages pass averages while worst-decile sentences sit at retrieval failure; the tail is *semantic* (anaphora, culturally-bound vocabulary), not entity noise; **Alignment-at-Risk (AaR)** reports it.
7. **Confounder-adjusted validity**: on this grid, ~ 40 – 60% of the raw metric $\leftrightarrow$ performance correlation is observables (data size, family, script, fertility); the surviving range is $\rho \approx 0.19$ – 0.31 (the July figure of about 0.59 on a benchmark-capable subset is withdrawn: not reproducible from any stored artifact). The tail measure AaR adds the tail visibility that is its job; MEXA is reported alongside for reference and is not ranked against it.
8. **Predicts behavior beyond observables (observationally)**: alignment at the LFS-identified layer predicts generation-time cross-lingual context use at confounder-adjusted $\rho = 0.67$ (language-clustered CI [0.55, 0.77]; mismatched-context controlled). Tested under intervention, the association does not transfer: see the Aim-2 entry below.
9. **Prospective test, 4/4 blind**: two untouched families (SmolLM2, Falcon3) matched their predicted recipe class; held-out validity exceeded the development CI; preregistered prediction P4 (a tail-measure-versus-MEXA difference on the benchmark-capable subgroup) passed on one held-out model ($\Delta = +0.14$, CI [+0.03, +0.26]) and is not used as a ranking.
10. **Aim-2 study, 28 conditions $\times$ three seeds, four objective families**: every condition fails its frozen criteria. The objective §3.2.2 degenerates by **89%**, and *neither LFS nor MEXA detects it*: both are scale-invariant, and the scale-collapsed condition posts among the highest retrieval and best tail alignment in the study. The decomposition separates collapsed from sound by 20 $\times$; the ratio divides that out. **Report the component profile, not the share**. A collapse guard is necessary and structurally insufficient (bounded hinge, unbounded objective). *LFS is a measurement, not a training target*.

How it was validated: every main claim pre-registered or adjusted for observables; failures printed beside successes; code and frozen protocols release-ready. **Ask**: 30 minutes; (i) the constrained Aim-2 objective the study implies, (ii) venue + remaining pre-submission items, (iii) coordination with adjacent work in the group.